Jiaao (Oliver) Lyu

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Summary

M.S. Computer Science student with experience in Linux kernel networking and backend development using Java and Spring Boot. Implemented a BBR-based slow-start extension in the Linux kernel and co-authored an ACM IMC 2026 paper. Seeking Software Engineer roles in backend or systems development.

Education

University of Pittsburgh — Pittsburgh, PA
M.S. in Computer Science | GPA: 3.96/4.0 — Aug 2025 - May 2027 (Expected)

Jiangsu University — Zhenjiang, China
B.E. in Intelligence Science and Technology — Sep 2021 - Jun 2025

Skills

Languages: Java, Kotlin, Python, C, JavaScript, SQL, C#
Frameworks: Spring Boot, Spring Security, React, Ktor, Jetpack Compose, Room, Retrofit, Hilt
Data & Cloud: PostgreSQL (PostGIS), MySQL, Docker, AWS (ECS, ECR, RDS, Amplify), Google Cloud (Cloud Run, Cloud SQL, Cloud Storage)
Systems & Tools: Linux kernel networking (TCP congestion control), Git, Linux

Experience

University of Pittsburgh | SUSS Replication and Extension — Nov 2025 - Sep 2026
Graduate Researcher | *Advisor: Asst. Prof. Amy Babay* | Research Code

- Reproduced SUSS on Linux 6.8.12 across six AWS paths and one residential Wi-Fi path, testing 24 flow sizes from 0.5 to 12 MB with 20 trials per condition.
- Implemented BBR-SUSS in C with per-connection state, bandwidth-growth and RTT safety checks, runtime controls, and a 25% pacing and cwnd gain boost during selected BBR startup rounds. Reduced flow completion time by about 1.4-9.1% versus BBR for 0.5-12 MB flows on tested paths with RTT > 50 ms.
- Built Python tools to analyze and plot flow completion times across paths and flow sizes.
- Co-authored the ACM IMC 2026 paper on RTT sensitivity, path-specific variation, and deployment constraints; prepared the public research artifact.

Institute of Software, Chinese Academy of Sciences (ISCAS) — Jul 2024 - Aug 2024
Self-Driving Algorithm Intern

- Built a Unity-based autonomous-driving simulation with vehicle-control modules and a rear-view camera.
- Designed a finite-state machine for traffic-light phase transitions; tested lane following and responses to traffic signs.
- Built a Java dashboard to display vehicle speed and steering angle.

Projects

StayBooking | Java, Spring Boot, React, PostgreSQL (PostGIS) — May 2026 - Sep 2026
Frontend | Backend

- Built React interfaces for guests to search, book, and cancel stays and for hosts to publish listings and review reservations.
- Secured Spring Boot APIs with JWT authentication, guest/host role-based access, and BCrypt password hashing.
- Implemented PostGIS queries to filter listings by radius, guest capacity, and date availability; rejected bookings that overlap existing reservations.
- Deployed the backend to Google Cloud Run with Cloud SQL for PostgreSQL and the frontend to AWS Amplify.

Netflix | Kotlin, Jetpack Compose, Ktor — May 2026 - Sep 2026
Frontend | Backend

- Built an Android app for browsing content, viewing video details and episodes, managing profiles, and playing video.
- Used MVVM, repositories, Compose Navigation, Hilt, and Retrofit; stored favorites with Room and implemented fullscreen playback with Media3 ExoPlayer.
- Built Ktor endpoints for video details, recently viewed history, and local media files.

Publications

A. Gatti, **J. Lyu**, T. Deshpande, A. Babay. Replicating and Extending SUSS: TCP Slow-Start Acceleration on AWS. *ACM IMC 2026.*
S. Cai, Y. Zhao, **J. Lyu**, et al. DDP-DAR: Network Intrusion Detection Based on Denoising Diffusion Probabilistic Model and Dual-Attention Residual Network. *Neural Networks, vol. 184.*